BEGINNING WITH COPERNICUS in the middle sixteenth century, the issue was formally joined. The picture of the Sun rather than the Earth at the center of the Universe was understood to be dangerous. Obligingly, many scholars were quick to assure the religious hierarchy that this newfangled hypothesis represented no serious challenge to conventional wisdom. In a kind of split-brain compromise, the Sun-centered system was treated as a mere computational convenience, not an astronomical reality—that is, the Earth was *really* at the center of the Universe, as everybody knew; but if you wished to predict where Jupiter would be on the second Tuesday of November the year after next, you were permitted to pretend that the Sun was at the center. Then you could calculate away and not affront the Authorities.*

"This has no danger in it," wrote Robert Cardinal Bellarmine, the foremost Vatican theologian in the early seventeenth century,

> and suffices for the mathematicians. But, to affirm that the Sun is really fixed in the center of the heavens and that the Earth revolves very swiftly around the Sun is a dangerous thing, not only irritating the theologians and philosophers, but injuring our holy faith and making the sacred scripture false.

"Freedom of belief is pernicious," Bellarmine wrote on another occasion. "It is nothing but the freedom to be wrong."

Besides, if the Earth was going around the Sun, nearby stars should seem to move against the background of more distant stars as, every six months, we shift our perspective from one side of the Earth's orbit to the other. No such "annual parallax" had been found. The Copernicans argued that this was because the stars were extremely far away—maybe a million times more distant than the Earth is from the Sun. Perhaps better telescopes, in future times, would find an annual parallax. The geocentrists considered this a desperate attempt to save a flawed hypothesis, and ludicrous on the face of it.

When Galileo turned the first astronomical telescope to the sky, the tide began to turn. He discovered that Jupiter had a little retinue of moons circling it, the inner ones orbiting faster than the